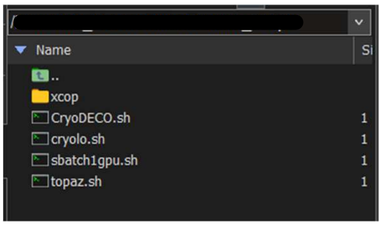
xcop Install Instructions

1. Put "xcop" in the location where you want to permanently store and keep



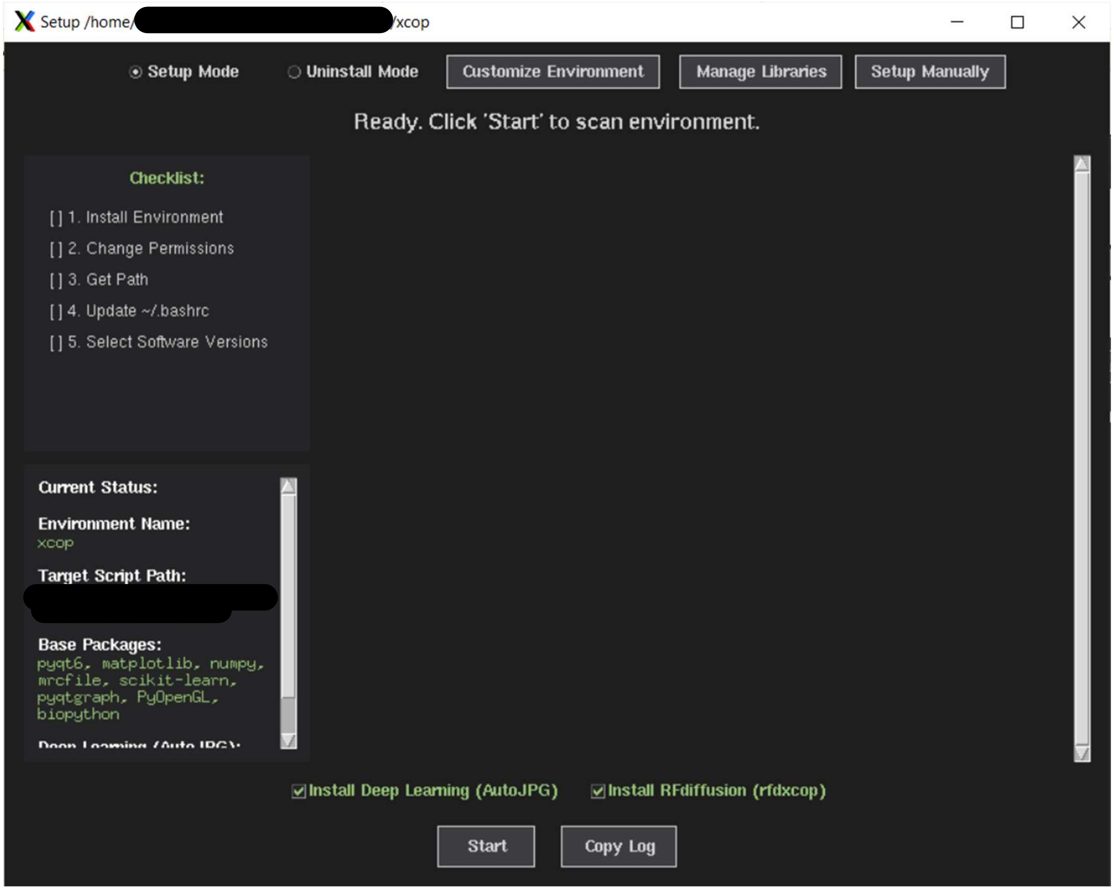
2. Go to the xcop folder

```
(base) [[redacted] xcop]$ cd /home/[redacted]/xcop/
```

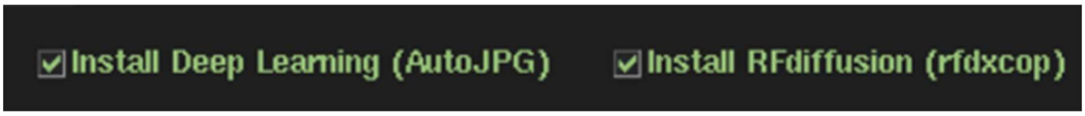
3. Run the xcop_setup.py script

```
(base) [[redacted] xcop]$ python xcop_setup.py
```

4. Click "Start" to start your installation

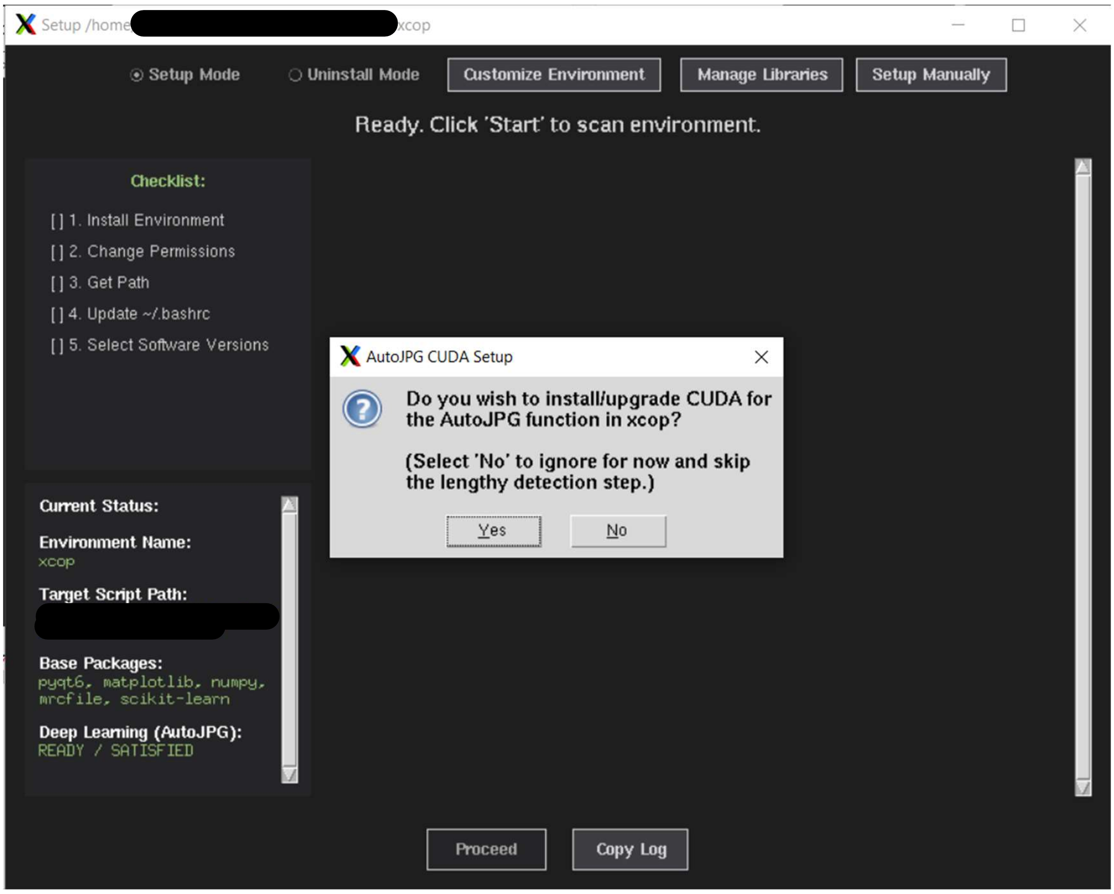


5. At the bottom you may select whether or not to install *AutoJPG* and *RFdiffusion*

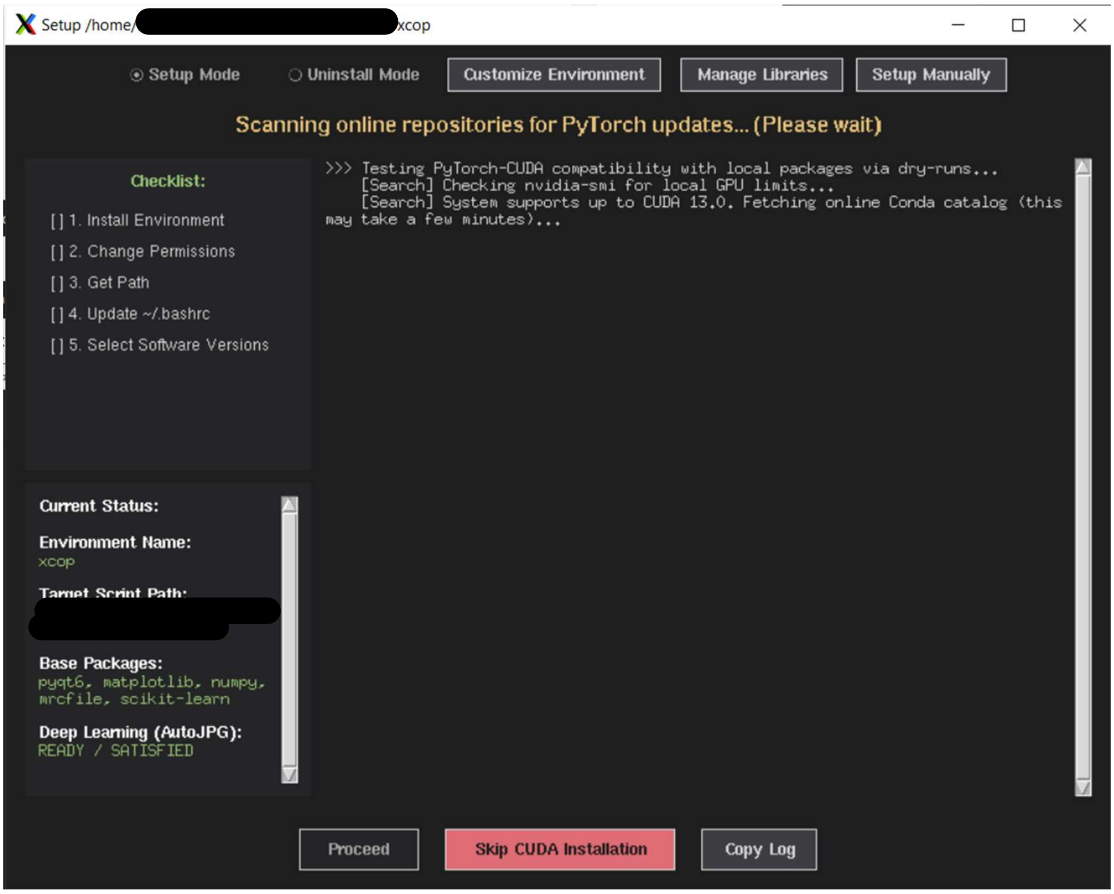


6. Choose "AutoJPG" for quickly removing empty micrographs

(This will take ~20 mins and will ONLY affect the "automated separation" of JPGs in xcop)



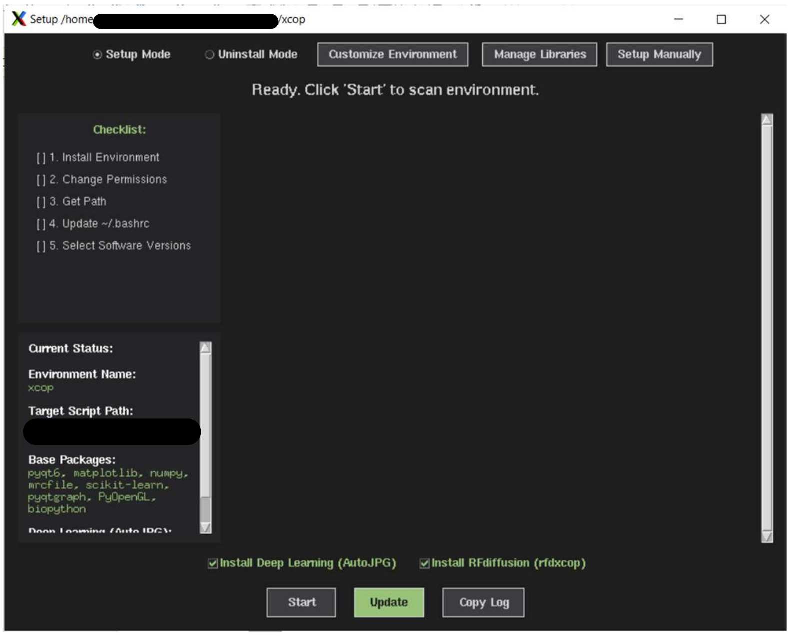
7. **Choose "RFdiffusion" if you wish to design binder for your proteins**
   (This will be installed as a slurm job so closing the setup program will not affect the installation for RFdiffusion once you started).
   However, you must NOT close the program until you finish setting up the xcop environment.
8. **It automatically detects and installs the environment for xcop scripts**



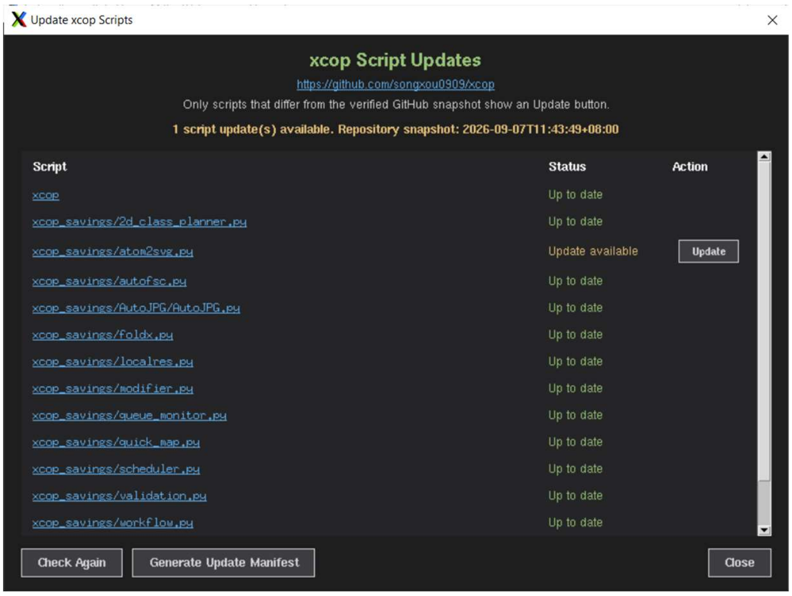
- Wait for the detection of the latest supported CUDA versions, you may also skip the lengthy version detection by clicking “Skip CUDA Installation”
  (This can take up to ~20 mins as it ensures mutual compatibility between CUDA and Python)

9. **A reminder will pop up to confirm the installation/upgrade of CUDA**
10. **Keep pressing “Proceed” for step 1, 2, 3**
11. **Step 4 is to add a path inside the bashrc file**
    - The purpose of this is to allow you to type “xcop” in the terminal to directly run xcop anywhere in the cluster
      (If you do NOT wish to change the bashrc, you may close the program now)
    - This modification is fully reversible when you uninstall the environment using this script (setup.py), the script can remove the lines (added by this script ONLY
) in the bashrc file
12. **Step 5 is to select the software versions installed on cluster**
    - This will be used by some functions in xcop (including RELION, XMIPP, SERVALCAT, PHENIX, CHIMERA), you may also do this in xcop
    - If the mentioned software is not installed, you may ask your IT support to install for you
13. **To use xcop: run** xcop& **anywhere**

14. To Update any scripts



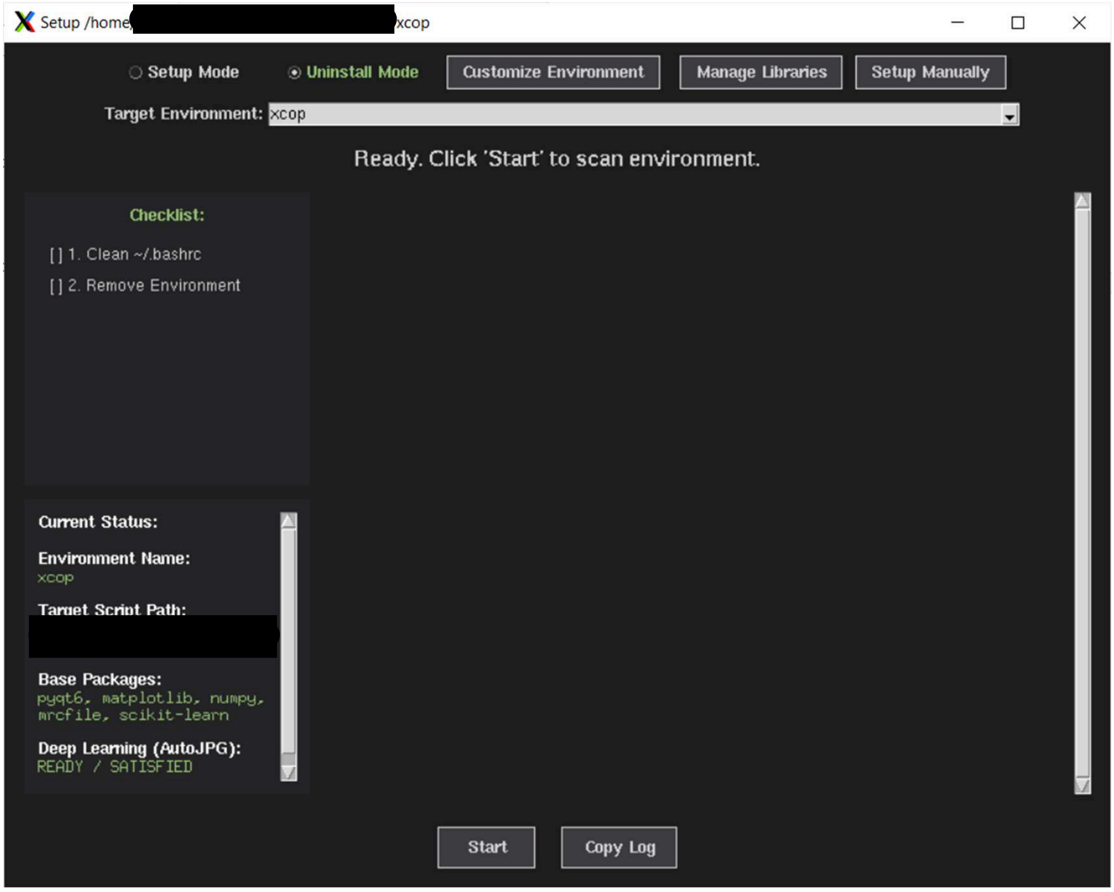
- Click the Update button at the bottom of the main window



- Click Update for the item you want to update

xcop Uninstall Instructions

1. Go to Uninstall Mode



2. Make sure the Target Environment is “xcop”, and click “Start”

- This script also allows you to install your own environment using the “Customize Environment”, but CUDA/PyTorch will not be detected or installed
- This script also allows you to manage installed python libraries using the “Manage Libraries”
- This script provides instructions for you to install the environment manually by typing commands in the terminal under the “Setup Manually” button